Week – I:

I. Initiation & Paper Breakdown

(To be completed after reading the paper's Abstract and Introduction)

- Abstract Summary:
- Problem Statement (According to the paper):
- Input / Output Specifications:
    - *Input Data:*
    - *Target Output:*
- Possible Implementations / Solutions (According to the paper):
- Possible Enhancements / Extensions Considered:
- Selected Enhancement for this Project:

II. Background / Literature Survey

(To be completed during the Related Work Deep Dive)

- Introduction & Domain Study:
- Technology Review (Libraries, Frameworks, & Tools):
- Historical Analysis & Timelines:
- Related Works Mapped (5–8 key papers):

III. Practical Implementation & Verification Checklist

(Hands-on execution log for Week 1)

- [ ] Audio Fundamentals & Playboard Setup: Environment installed (librosa, torchaudio), sample WAV loaded, waveform plotted, sample rates tested.
- [ ] Audio Representation & Display: Computed STFT, Mel-spectrogram, and CQT; experimented with n_fft and hop_length.
- [ ] SDLC & Repository Setup: Initialized GitHub structure (data/, src/, notebooks/, results/, docs/) and tracking board.
- [ ] Bulk Data Pipeline: Wrote scripts to index, validate, check durations, and segment 50+ audio files.
- [ ] Feature Exploration & Similarity: Computed cosine similarity/DTW matrices and inspected embeddings via t-SNE.